

Sean Coath

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EDUCATION

University of Notre Dame

Notre Dame, IN | May 2029

Bachelor of Arts 3.97 / 4.00 (Dean's List All Semesters)

Degrees/Programs: Computer Science Major (AI & Data Science Concentration), Philosophy Major (Philosophy of Science & Mathematics Concentration), Mathematics Minor, Glynn Family Honors Program

Activities: ND Symphony Orchestra (Principal Tuba), ND Marching Band (Sousaphone)

Notre Dame College Prep

Niles, IL | May 2025

High School Diploma 5.36 / 4.00 (Summa Cum Laude)

Honors: Valedictorian, National Merit Commended Scholar, AP Scholar with Distinction, IHSA Scholastic Award

COMPUTING EXPERIENCE

Data Inference Analytics & Learning (DIAL) Lab

Notre Dame, IN | August 2026

RISE AI Developer

- Will begin research in developing the RISE (Responsible, Inclusive, Safe, & Empowering) AI platform under Dr. Nitesh Chawla in Notre Dame's Lucy Institute

Handshake AI Fellowship

Remote | May 2026 – present

Video Reinforcement Learning & Application Integration AI Training Specialist

- Develop & evaluate domain-specific prompts to assess lingering performance weaknesses of LLMs
- Analyze LLM outputs for scientific accuracy, clarity, & depth in specialized subfields
- Completed over 100 approved tasks across various projects & promoted to Task Reviewer for strong performance

University of Notre Dame Department of History

Remote | May 2026 – present

Paid Research Assistant & Technical Developer

- Building an AI-powered GIS visualization of world history to be used for an upcoming ND course in Spring 2027
- Writing Python scripts with various libraries to scrape & geocode historical data sources into a refined database
- Architecting a web-based framework to deploy visualizations with a user-friendly UI using HTML & JS

University of Notre Dame Center for Research Computing

Notre Dame, IN | Summer 2024

Research Computing Scholar (Pre-College Program)

- Designed technical architecture & business proposal for an AI-driven college recommendation service
- Researched ethical constraints of student data privacy when analyzing open-source educational datasets

INDEPENDENT PROJECTS

Jeopardy!-Style Quiz Game | Python, Pygame, gTTS

- Leveraged Pygame for graphics, sound effects, AI-generated voice playback, & procedural animation
- Scraped ~500,000 clues & restructured into a JSON dataset for efficient filtration & cleaning based on metadata

Golf Simulation Engine | C, X11, Linux

- Engineered a custom golf physics engine & dynamic level editor featuring interactive obstacle placement

Additional Information & Projects at seancoa.github.io

SKILLS

Relevant Coursework: Data Structures, Systems Programming, Discrete Mathematics, Linear Algebra / Differential Equations, Fundamentals of Computing, Calculus III

Certifications: LPI Linux Essentials (in progress), AWS Certified Cloud Practitioner (in progress)

Technical Proficiencies: Python, C, HTML, Linux

Foreign Language: Intermediate Spanish (AP Spanish Language & Culture Score: 5)

OTHER PROFESSIONAL EXPERIENCE

Various Tutoring Companies

Chicago, IL | June 2026 – present

ACT / SAT / HSPT Tutor

- Conduct classes and one-on-one sessions to teach students optimal test-taking strategies & provide supplemental instruction for challenging topics

Self-Employed

Skokie, IL | Summer 2023 – Summer 2025

Landscape Maintenance Provider

- Managed a consistent work schedule & accurate records without direct supervision, resulting in 100% customer retention

SERVICE

Computer Science Mentor / Tutor, CovEducation

Chicago, IL | 2025 – present